Computer Science, along with PhD student, Jon Oberheide and postdoctoral fellow Evan Cooke, evaluated 12 different antivirus software programs (including the popular McAfee, Avast and AVG) by pitting them against more than seven thousand malware samples. They found that, due to the increasingly innovative viruses and the growing complexity of anti-virus software itself, detection of malicious software was really low — about 35 per cent. Besides having several vulnerabilities in the software itself, most of them took about seven weeks on an average to equip themselves against new virus threats that are in circulation on the Web.

Another major drawback in today's anti-virus packages is that you can't run more than one of them simultaneously in the same system. CloudAV is a single solution to all of these problems for the following reasons:

- It analyses potential threats to your system using several different antivirus programs at the same time, thereby significantly increasing the degree of protection for your system.
- Operates by installing a simple, lightweight software agent in your computer, mobile phone or laptop, which automatically detects a new document or program being opened and sends it to the anti-virus cloud (somewhere on the web) to be analysed.
- With CloudAV it's pouring antivirus agents. CloudAV uses 12 different detectors that run parallel to one another, but independently, to tell your computer whether it's okay to open a particular file.
- Caches the results so that detection becomes smoother in future.

And If It Comes Through...

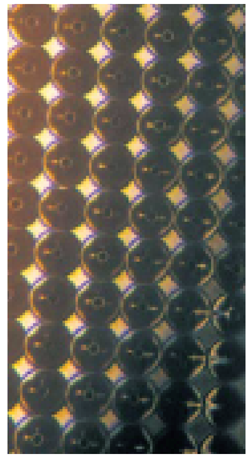
The latest irritant in India is frequent virus attacks on our cell phones. Typically, cell phones lack the space and power to accommodate bulky antivirus software. Leaving the job of detection and quarantine to an external agent — and not just one, but twelve — would be a boon for users of mobile computing devices. For the rest of us too — PC users — we'll stop cursing our favourite AV vendor for the viruses that weasel in and start praising CloudAV.

5. TELESCOPIC PIXELS

Mirror Writing

Screening For More

As alternatives to CRTs are becoming cheaper, more than half the globe has switched over to LCD monitors or TVs — not to mention the ubiquitous TFT screens in our cell–phones and PDAs. Naturally, we have also begun to perceive the manifest errors and glitches in using LCD technology. Not to rest on their laurels, scientists have already begun investigating possible new technologies to replace LCD screens. And this time, they are doing it with mirrors.



Pixel this!

Whether it's LCD, Plasma or CRT screens, we're stuck with pixels. Pixels — short for 'Picture Elements' — are the tiny dots that make up the images on a screen. To cut a long story short, the quality and accuracy of the image is determined by the 'resolution' of the screen, so the greater the number of pixels, the sharper the image.

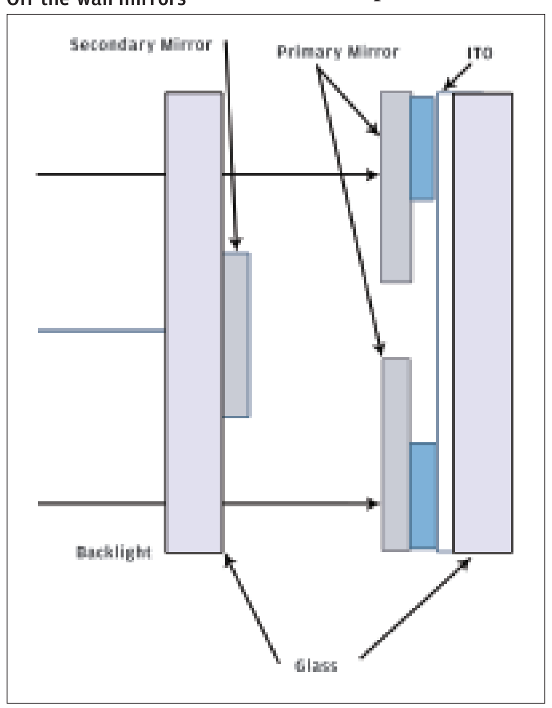
Even though LCD screens are all the rage, there are several drawbacks that noticeably hinder the achievement of a truly high-quality image:

- The pixels in an LCD screen do not really turn completely off.
- It's virtually impossible to view the image on a TFT/LCD screen in natural ambient light.
- When images move fast, the pixels take about half a second to switch between colours, and when these are very different, this leads to momentary blurs.
- Dead or stuck pixels which are damaged in such a way that they permanently stay in the or off state, seriously affect visual accuracy.
- Finally, by the time light passes through the three stages of an LCD screen (the polarising film, the liquid–crystal coat and the colour filters) almost 90 per cent of the light is lost, making the screen appear blacker and the displayed image dark.

Microsoft To The Rescue

Researchers at Microsoft have come up with a terrific new design for pixels (published online in Nature Photonics, 20th July, 2008) in which each individual pixel is made up of two opposing microscopic mirrors with one changing shape based on applied voltage, and reflecting light through a hole on the primary mirror and onto the display screen. Both mirrors are made of aluminium and the first one, with a hole in the center, is only a 100 micrometres wide and 100 nanometres thick!

Off-the-wall mirrors



When the pixels are 'off', both mirrors reflect the light back on to the source, so none emerges on the other side of the screen. However, when they're switched on, the disk bends towards a transparent electrode (typically made of indium titanium oxide) due to a little application of voltage. The light therefore bounces towards the second mirror and emerges through the hole.

And If This Comes Through...

Michael Sinclair, senior researcher in the Hardware Devices group — under the direction of Turner Whitted at Microsoft — is convinced that once the design makes it past the prototype stage, it will replace conventional display units all over the world. Less powerful backlights would be necessary and this would bring down costs, while increasing the longevity of the battery on your cell phone or laptop. The telescopic pixels allow about 36 per cent of the light through, increasing brightness by three to six times as compared with the present-day LCD technology.

Just as happened with CRT monitors, people are going to sooner or later hope to get some more space to use on their shrinking office desks and the Telescopic Pixel technology could shrink the width of the screen to the thinness of a whiteboard. As the design is simple and the materials are cheaper, the fabrication as well as price should be substantially easier on the pocket. One possible drawback could be the mechanical nature of the parts — mechanical parts tend to wear out and break down — which may raise maintenance issues, but the positives far outweigh this single danger. So, though we're not holding our breath, we're definitely looking forward to the quick development and commercialisation of the Telescopic Pixel Screen.

6. SENSITIVE ARTIFICIAL LISTENERS

Sensitising Machines

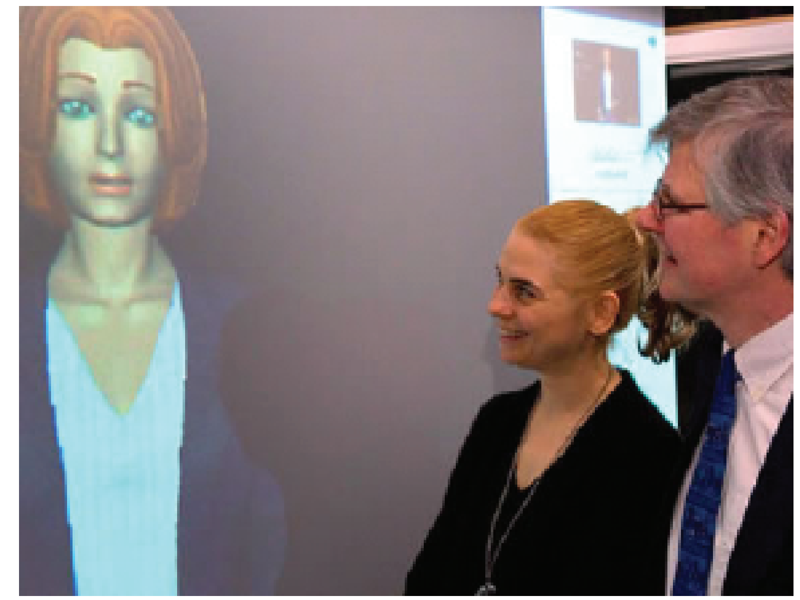
More Like People

We're already quite familiar with our computers interacting through auditory means — voice commands, text-to-speech software, etc. — but most of us get a bit bugged by the monotony of the electronic voice speaking back to us. Science fiction writers have always dreamt of computers becoming emotional or talking like people (there was even this computer which fell in love in the 1984 Hollywood movie 'Electric Dreams'). The fiction may be inching towards fact with the Sensitive Artificial Listener system (SAL) being developed by an international team including Queen's University, Belfast.

Making Human Inputs More Acceptable To Machines

Humans do not only communicate through words. Non-verbal communication, in fact, is supposed to constitute more than 90 per cent of our oral interactions. Computers, however, only understand crystal clear commands and the ambiguity, fluidity and variant significations of body language or facial expressions has so far been beyond machines.

Using a unique blend of science, ethics, psychology and linguistics, scientists are attempting to overcome this obstacle too. SEMAINE (Sustained Emotionally colored Machine-human Interaction using Nonverbal Expression) is a project undertaken by an international group of technologists led by DFKI, the German centre for research on Artificial Intelligence and including Imperial College, London, the University of Paris, the University of Twente in Holland, Queen's University, Belfast and the Technical University of Munich. The team, with a European Commission Grant of 2.75 million euros, aims to create a Sensitive Artificial Listener (SAL) system, which will perceive a human user's facial expression, gaze, and voice while interacting with him or her. Just as humans do, the system will alter it's own tone, behaviour and actions according to the non-verbal stimulus it receives (and actually perceives) from the user. For the first time in history, this project to create a machine-human interface system is using fields as diverse as psychology, linguistics and ethics at every step of its endeavour.



GRETA — an 'embodied conversational agent' at Queen's University

And If This Comes Through...

Professor Roddy Cowie, from the School of Psychology, at Queen's University gives a timeline of about 20 years. But given the scale and enthusiasm of the SEMAINE project, and given that several such projects are underway all over